

Frequencies and Channels

At the end of this episode, I will be able to:

1. Frequencies and Channels

Leamer Objective: *Learn about the wireless frequencies and channels used in modern wireless LANs*

Description: In this episode, you will learn about the wireless frequencies used in networking today. You will also learn about the channels used in wireless technologies. This includes a discussion of channel width, non-overlapping channels, and 802.11h.

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- 802.11 protocols define portions of the RF spectrum for wireless
 - The standards define several distinct radio bands for WiFi:
 - 860/900 MHz
 - 2.4 GHz
 - 3.6 GHz
 - 4.9 GHz
 - 5 GHz
 - 5.9 GHz
 - 6 GHz
 - 45 GHz
 - 60 GHz
 - Channels are numbered at 5 MHz spacing within a band (except in the 45/60 GHz band, where they are 0.54/1.08/2.16 GHz apart) and the number linearly relates to the center frequency of the channel.
 - Although channels are numbered at 5 MHz spacing, transmitters generally occupy at least 20 MHz, and standards allow for channels to be bonded together to form wider channels for faster throughput.
 - 802.11h is intended to resolve interference issues introduced by the use of 802.11a in some locations, particularly with military Radar systems and medical devices. Dynamic Frequency Selection (DFS) detects the presence of other devices on a channel and automatically switches the network to another channel if and when such signals are detected. Transmit Power Control (TPC) reduces the radio frequency (RF) output power of each network transmitter to a level that minimizes the risk of interference.
 - Most WiFi modems come with two frequencies – 2.4 GHz and 5 GHz – and ask you to choose between them when connecting a device to your home network. This is known as dual band.
 - Band steering automatically connects your devices to the best available WiFi frequency in your home.
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Additional Resources:

List of WLAN Channels

https://en.wikipedia.org/wiki/List_of_WLAN_channels